Types of Forces

Part A

Materials

- 20 cent coin
- Small square of card (approximately the same size as coin)



Method

1. Hold the piece of card in one hand and the coin in the other, at the same height.
2. Drop them both at the same time.

Observations

1. Describe what happened to the card and the coin.

2. What force(s) were acting on the card and the coin?

3. Is the force(s) acting on the card and the coin a contact force or a non-contact force?

4. Are the forces acting on the card and the coin balanced or unbalanced? Give a reason for your choice.

Part B

Materials

- 20 cent coin
- Wooden peg
- Small square of card



Method

1. Stand the peg on one end and place the card on top of the peg. Balance the coin on the card.
2. Holding the peg at the base, try to flick the card out from under the coin, keeping the coin balanced on top of the peg.

Observations

1. Describe what happened to the coin, the card and the peg.

2. What type of force acted on the card when you flicked it?

3. Why could you flick the card out from under the coin and leave it on the peg?

4. Are the forces acting on the card balanced or unbalanced? Give a reason for your choice.

5. Are the forces acting on the coin balanced or unbalanced? Give a reason for your choice.

Part C

Materials

- 2 bar magnets with their poles marked



Method

1. Hold one magnet in each hand with opposite poles facing towards each other.
2. Slowly bring them closer together.
3. Now, hold one magnet in each hand with the same poles facing towards each other.
4. Slowly bring them closer together.

Observations

1. What did you notice when the opposite poles of the magnet were facing towards each other?

2. What did you notice when the same poles of the magnet were facing towards each other?

3. What type of force is acting between the magnets?

4. Is this force a contact or a non-contact force?

Part D

Materials

- Plastic pen
- Small pieces of paper

Method

1. Rub the plastic pen quickly on your clothes.
2. Bring the pen close to the pieces of paper.

Observations

1. What happens when you bring the pen near the pieces of paper?

2. The force here is due to an electrostatic charge on the pen. Does the pen need to be touching the paper or can the electric force act over a distance?



3. Is an electrostatic force a contact or non-contact force?

Part E

Materials

- Table tennis ball
- Drinking straws

Method

1. Gently roll the table tennis ball towards you.
2. Use the straw to blow on the ball as it moves towards you.
3. Record your observations below.
4. Gently roll the table tennis ball away from you.
5. Use the straw to blow on the ball as it moves away from you.
6. Record your observations below.
7. Gently roll the table tennis ball across the table in front of you.
8. Use the straw to blow on the ball as it move in front of you.
9. Record your observations below.

Observations

1. Describe what happened when you blew on the table tennis ball while it moved towards you.

2. Describe what happened when you blew on the table tennis ball while it moved away from you.

3. Describe what happened when you blew on the table tennis ball while it moved across in front of you.

4. Is this an example of a contact force or a non-contact force? Give a reason for your choice.

Extra for Early Finishers: Tabletop Soccer (2 or 4 players)

Materials

- Table tennis ball
- Drinking straws
- Blu-tack

Method

1. Using the blu-tack, make two sets of goals by sticking three straws together in a U-shape.
2. Blu-tack the goals to the desk at either end.
3. Team members stand at either end of the desk and try to score a goal by blowing the table tennis ball across the table.